An explicit genus-3 plane quartic over F_q for an elliptic curve over F_{q³}

2026-10-12. Conductor T-2267. **Structural stage diagnostic on toy fields.** A curve is built and its Frobenius checked. No discrete logarithm is solved, no index calculus is run, and no speedup is claimed. Protocol frozen before the first run: PROTOCOL.md. Run `20261013T0320Z`: records, environment, run report. Addendum run `20261013T0347Z` (the order-332 class at q = 7, see below): records, environment, run report. This report as PDF: REPORT.pdf. Follows `research/ghs_cubic_cover_census_20261012/` (PR #1759), which counted these covers by genus without building them.

Question

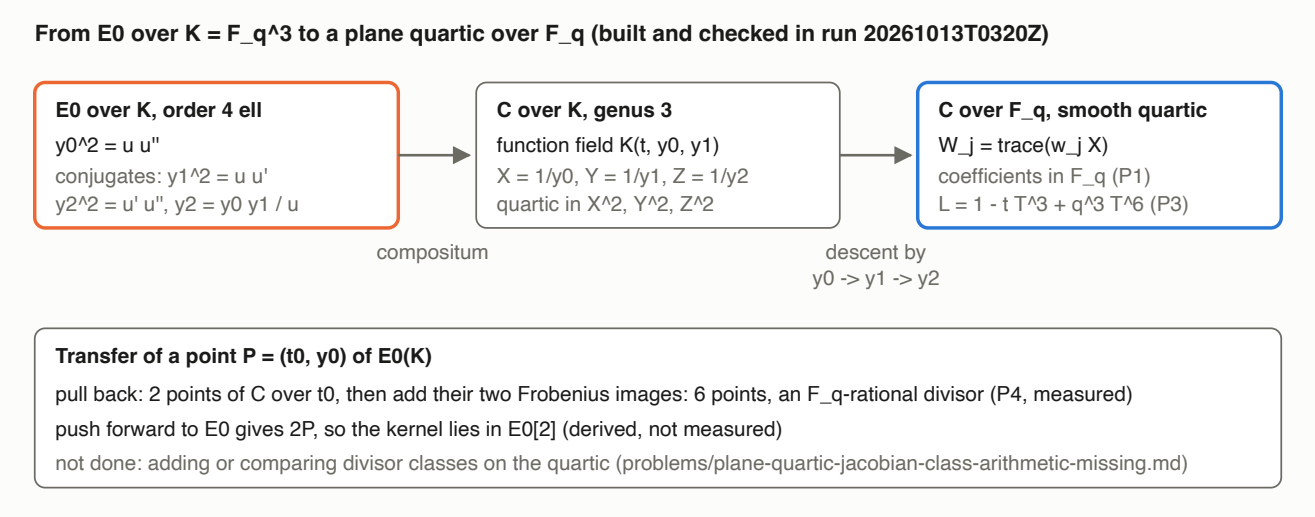
The census found that every isogeny class with `4 | #E(K)` over `K = F_{q³}` has a genus-3 cover by branch data. Does that cover exist as a curve over `F_q` whose Jacobian has `#E(K)` rational points, and is the transfer of points an explicit `F_q`-rational object?

Result

Yes on all eleven cases built (`q ∈ {5, 7, 11}`, `#E0(K) = 4ℓ`; nine in the first run, the two of the order-332 class at q = 7 in the addendum). For each:

- the curve is a **smooth plane quartic with coefficients in `F_q`**, so it is not hyperelliptic;
- its point counts over `F_q`, `F_{q²}`, `F_{q³}` give L-polynomial `1 − t_E T³ + q³T⁶`, hence **`#Jac(C)(F_q) = #E0(K)`**;
- every transfer divisor tested is an `F_q`-rational divisor on it.

So the census of PR #1759 counted objects that exist as assumed. What is still not shown by execution is the transfer as a group homomorphism, because no class arithmetic on a plane quartic is implemented (problem record).



Construction (derived; two steps in Lean)

`u ∈ K[t]` monic quadratic, `u' = u^σ`, `u'' = u^{σ²}`, six distinct roots, `u, u', u''` linearly independent.

1. `E0 : y0² = u·u''`, `E1 = E0^σ : y1² = u·u'`, `E2 : y2² = u'·u''` with `y2 = y0·y1/u` (`DescentCycle.third_cover`).
2. `σ~ : y0 ↦ y1 ↦ y2 ↦ y0`, `t ↦ t`, σ on K, has order 3 (`DescentCycle.cycle_closes`), so the compositum descends to `F_q`.
3. With `X = 1/y0, Y = 1/y1, Z = 1/y2`: `(u : u' : u'') = (Z² : X² : Y²)`. If `(1, t, t²)ᵀ = R·(u, u', u'')ᵀ` then `t·t = 1·t²` reads `G = (R₁·w)² − (R₀·w)(R₂·w) = 0, w = (Z², X², Y²)`: a quartic in `X², Y², Z²`. The three differentials `dt/y_i` are holomorphic and `(X : Y : Z)` is the canonical map, so a smooth `G` means `C` is not hyperelliptic.
4. σ~ permutes `X → Y → Z`. In `W_j = ω_j X + ω_j^σ Y + ω_j^{σ²} Z`, `ω = (1, g, g²)`, the quartic has `F_q` coefficients up to a scalar.
5. Transfer `φ = (1 + Fr + Fr²)∘π0^*`, with `Fr(t, y0, y1, y2) = (t^q, y2^q, y0^q, y1^q)`. `π_j_* ∘ π_0^* = 0` for `j ≠ 0` (it factors through the t-line), so `π0_* ∘ φ = [2]` and `ker φ ⊂ E0[2]`. With `#Jac(C)(F_q) = #E0(K) = 4ℓ`, φ maps `E0(K)[ℓ]` isomorphically onto `Jac(C)(F_q)[ℓ]`.

Report statement	Lean declaration	Status
`(u·y2)² = u²·u'·u''`	`third_cover`	proved, any commutative multiplication
`u·(y1·y2) = (u·u')·y0` (cycle closes)	`cycle_closes`	proved
cancelling u; the function field; descent; step 5	none	**not formalized**

Lean: lean/DescentCycle.lean, toolchain v4.34.1, no Mathlib, no axioms. No identity certificate: the report's identities are conditional on the curve equations, not polynomial identities.

Measurements (run 20261013T0320Z, exact counts)

Tool: genus3_build.rs. `u = t² + bt + c` in lexicographic order of field index; the first cases with `#E0(K) = 4ℓ`. `N_k` are projective point counts of the `F_q` quartic over `F_{q^k}` by exhaustion; `N3` was also counted on the `K` model and agreed in all nine cases.

q	u (index of b, c)	roots of u in K	#E0(K)	t_E	in F_q	smooth	N1, N2, N3	predicted	#Jac(C)(F_q)
5	5, 30	0	148 = 4·37	−22	yes	yes	6, 26, 192	6, 26, 192	148
5	5, 51	2	148 = 4·37	−22	yes	yes	6, 26, 192	6, 26, 192	148
7	7, 52	0	356 = 4·89	−12	yes	yes	8, 50, 380	8, 50, 380	356
7	7, 54	0	356 = 4·89	−12	yes	yes	8, 50, 380	8, 50, 380	356
7	7, 110	2	356 = 4·89	−12	yes	yes	8, 50, 380	8, 50, 380	356
7	7, 118	2	356 = 4·89	−12	yes	yes	8, 50, 380	8, 50, 380	356
7	7, 129	2	356 = 4·89	−12	yes	yes	8, 50, 380	8, 50, 380	356
11	11, 130	0	1348 = 4·337	−16	yes	yes	12, 122, 1380	12, 122, 1380	1348
11	11, 172	2	1324 = 4·331	8	yes	yes	12, 122, 1308	12, 122, 1308	1324
7†	49, 8	0	332 = 4·83	12	yes	yes	8, 50, 308	8, 50, 308	332
7†	49, 9	0	332 = 4·83	12	yes	yes	8, 50, 308	8, 50, 308	332

† Addendum run `20261013T0347Z`: the second 4ℓ isogeny class at `q = 7` (`#E0(K) = 332`, trace +12, the sign opposite to the 356 class's −12). The frozen first-in-index rule of the first run reached only the 356 class. Exhausting the parametrization `u = t² + bt + c`, this class has exactly two irreducible-u representatives satisfying the genus-3 conditions and **no split-u representative** (65432 candidates examined). The order-356 run was not exhaustive, so this is not a multiplicity comparison between the two classes.

Example, `q = 7`, u index (7, 110), over `F_7`:

```
W2⁴ + W1²W2² + 6W1³W2 + 6W1⁴ + 4W0W2³ + W0W1W2² + 4W0W1²W2 + 4W0W1³
  + W0²W2² + 4W0²W1W2 + 5W0²W1² + 6W0³W2 + 6W0⁴ = 0
```

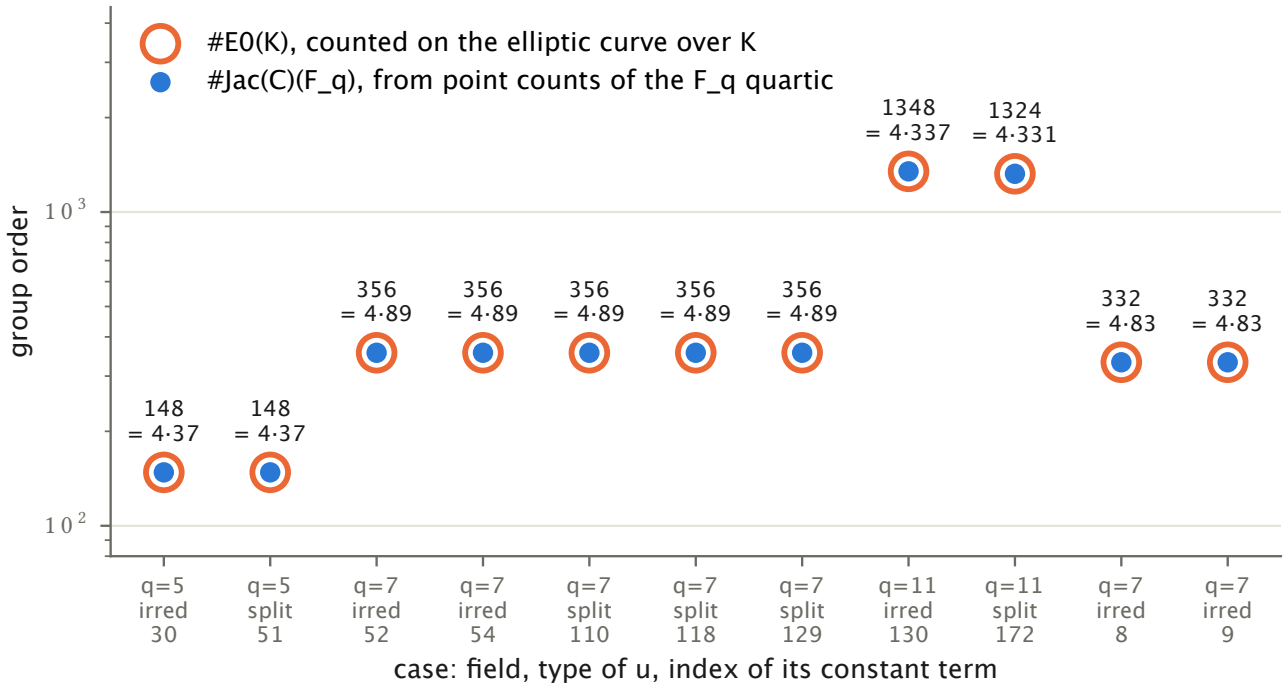
All eleven quartics are in the records (`quartic_Fq`, entries `[i, j, k, coefficient]` for `W0^i W1^j W2^k`; the two order-332 quartics are in the addendum run's records).

Transfer divisors (P4). For each `P = (t0, y0) ∈ E0(K)` with `u·u'(t0)` a non-zero square, the two points of `C` over `P` and their two Frobenius images each:

q	u	points tested	on the F_q quartic	q-power = model Fr	image on E0	points of E0 not tested
5	5, 30	564	564	564	564	52
5	5, 51	564	564	564	564	48
7	7, 52 / 7, 54	1128 each	1128	1128	1128	166 each
7	7, 110 / 118 / 129	1092 each	1092	1092	1092	168 each
11	11, 130	4128	4128	4128	4128	658
11	11, 172	3876	3876	3876	3876	672
7†	49, 8 / 49, 9	912 each	912	912	912	178 each

† Addendum run `20261013T0347Z` (order-332 class).

The untested points are those where `u·u'(t0)` is not a square: their two preimages are conjugate over `F_{q⁶}` and need a field the tool does not build.



All eleven cases (nine from the first run, two order-332 from the addendum); the run-20261013T0320Z report keeps the nine-case graph. P1, P2, P3 and P4 hold on every case; no falsifier fired.

Scope and limits

- **Not executed:** φ as a homomorphism, its injectivity on `E0[ℓ]`, any discrete logarithm, any relation collection. These need class arithmetic on the quartic; see the problem record.
- **Selection.** The frozen rule takes the first cases in index order. All five `q = 7` cases of the first run landed in one isogeny class (`#E0(K) = 356`); the other 4ℓ class at `q = 7` (332) was built in the addendum run after a target-order argument was added to the tool (the argument defaults to off, so the first run reproduces byte for byte). Most candidates were skipped as linearly dependent (2394 of 2443 at `q = 7`) because `b ∈ F_q` comes first in index order.
- **One construction.** Only the `m = 2, r = 6` type. The hyperelliptic `m = 3, r = 5` type is not built.
- **Cost (published asymptotics, not measured):** index calculus on a smooth plane quartic over `F_q` is reported at about `q` operations (Diem), against about `q^{3/2}` for rho on `E(K)`. These toy fields are far too small to compare anything.
- `n = 3`, `q ∈ {5, 7, 11}`, odd characteristic above 3.

Next discriminating step

Implement linear equivalence on a plane quartic and execute the transfer on the `q = 7` case (7, 110); then relation collection by lines through pairs of `F_q`-points, counted against rho. The acceptance conditions are in the problem record.

Graphs checked and left unchanged

No canonical graph changes (no ECDLP rung, no timing; no `benchmarks/` entry).

Reproduce

```
rustc --edition 2021 -O genus3_build.rs && ./genus3_build 5 1 1 && ./genus3_build 7 3 2 && ./genus3_build
11 1 1
./genus3_build 7 3 2 332                # addendum: the order-332 class at q = 7
python3 reports/plot.py results/run-20261013T0320Z.jsonl reports/run-20261013T0320Z   # layout only
cat results/run-20261013T0320Z.jsonl results/run-20261013T0347Z.jsonl > results/all11.jsonl
python3 reports/plot.py results/all11.jsonl reports/run-20261013T0347Z   # then orders.* renamed to
orders_all11.*
cd lean && lake build
```

The fourth argument to `genus3_build` is an optional target order (0 or absent keeps the original first-in-index rule); `./genus3_build 7 3 2` reproduces the q = 7 records of run `20261013T0320Z` byte for byte.